Homework — 1.5.2 Moral and Ethical Issues — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. (a) [1] Any one: faster than manual reading / can process huge numbers of scans without fatigue / consistent results / acts as a safety net catching things a tired clinician might miss.

(b) [1] Any one named concern, e.g. **automated decision making / AI ethics** — the system lacks human judgement and could miss or wrongly flag a case, and it is unclear who is **liable** if it errs; or over-reliance could de-skill staff; or **bias** if trained on unrepresentative scan data.

2. (a) [2] 1 mark: AI learns patterns from its **training data**. 1 mark: if that data was **unrepresentative/biased** (e.g. far fewer examples of some groups), the model performs worse on / discriminates against those groups, **reproducing or amplifying** the bias.

(b) [1] The model is a "black box" — its internal reasoning is opaque, so it is hard to see *why* it made a decision, hard to detect/explain the bias, and hard to hold anyone accountable or appeal a wrong result.

3. [2] 1 mark benefit: lower labour costs / cheaper development / access to a wider talent pool / "follow the sun" round-the-clock work. 1 mark drawback: local job losses in the UK / quality, time-zone or communication difficulties / data-security and oversight concerns with work done abroad.

4. [2] 1 mark: a monopoly can mean high prices, vendor **lock-in** and stifled competition because customers depend on one supplier's proprietary product. 1 mark: open-source software is freely available to view, use, modify and share, so users are not locked to one vendor, can switch or self-host, and competition/standards are encouraged — countering the monopoly.

5. [5] Levels-of-response **"discuss"** question — mark the **whole answer**, not point-by-point. Expect a balanced discussion across stakeholders with a justified conclusion.

Indicative content — - **FOR:** cheaper to run / saves council money; faster, 24/7 access for the digitally-confident; less paperwork; easier to update. - **AGAINST:** the **digital divide** excludes those without internet access, devices or skills — disproportionately the elderly, poor, rural or disabled, who may be unable to claim essential benefits; loss of face-to-face help; widening inequality. - **Stakeholders (aim 3+):** the council/taxpayers · digitally-confident residents · digitally-excluded residents (elderly/poor/disabled) · staff whose roles change. - **Justified conclusion:** must take a clear, ideally *conditional* position, e.g. *"moving services online is justified to cut costs only if assisted/phone/in-person routes remain for those who cannot access them, so no one is excluded from essential services."*

Brief level descriptors: - **Level 1 (1):** one-sided / fragmented (e.g. only "saves money" or only "excludes the elderly"); little development; no conclusion. - **Level 2 (2–3):** some discussion of both sides but mostly descriptive or one-sided; limited stakeholders; weak or absent conclusion. - **Level 3 (4–5):** balanced, several stakeholders, **both** benefits and drawbacks with accurate terminology (e.g. *digital divide*) and sustained reasoning, leading to a **justified conclusion** that takes a reasoned position.

Part B (6 marks)

6. [2] 1 mark: Computer Misuse Act 1990. 1 mark: **Offence 2** — unauthorised access *with intent* to commit/facilitate a further offence (here, theft of trade secrets). *(Accept that Offence 1 also occurs, but Offence 2 is the more serious one and is required.)*

7. [2] 1 mark: **TCP** (Transport) splits data into packets, handles reliable delivery — sequencing, acknowledgement and reassembly of packets in the right order. 1 mark: **IP** (Network/Internet) handles addressing and **routing** of packets across networks from source to destination.

8. [2] 1 mark + 1 mark: a multi-core processor has more than one core, so it can execute **multiple threads/instructions truly simultaneously [1]**; tasks that can be split/parallelised are shared across cores and finish sooner (true parallelism rather than time-slicing on one core) **[1]**.

Total: 14 + 6 = 20 marks